Appendix. Baseline comparison of WPD against EMD/EEMD/VMD in separating the common and differential components

1.1 Comparison design and baseline settings

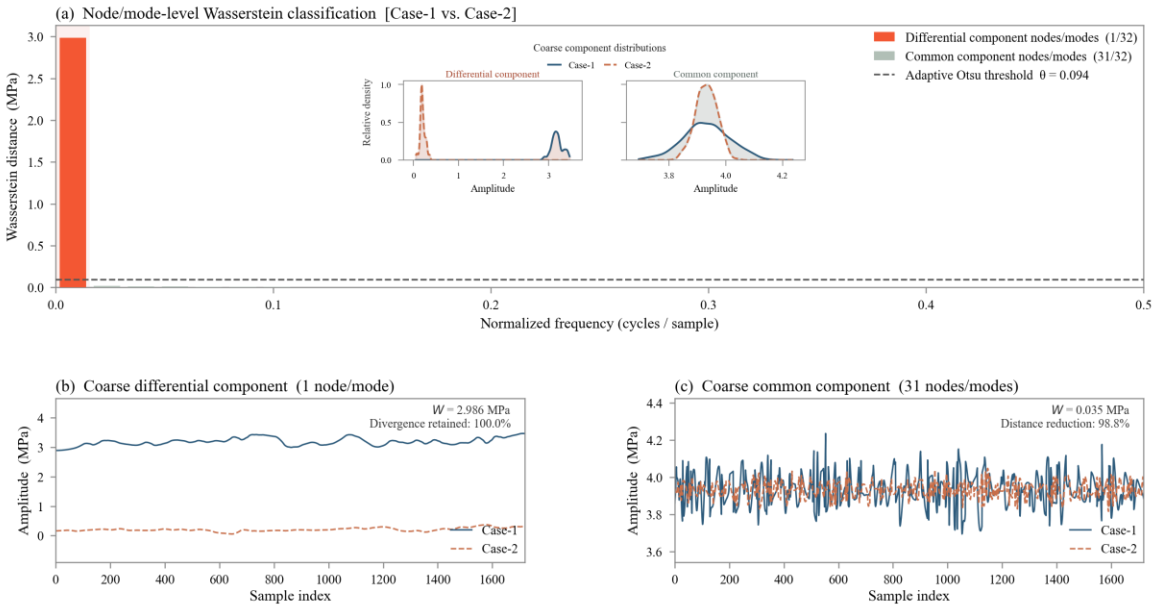
To examine whether the first stage of the two-stage funnel-shaped framework proposed in the main text (wavelet packet decomposition, WPD, with Wasserstein-distance discrimination of common and differential components) offers a discernible advantage over other decomposition paradigms that are equally widely used in signal processing, this appendix compares it systematically with EMD, EEMD and VMD on the same MWD data.

EMD, EEMD and VMD provide signal decomposition only and carry no mechanism for discriminating common from differential components; examined outside that discrimination logic, their raw decomposition results neither define nor allow any evaluation of separation performance, and so cannot serve directly as a baseline. This appendix therefore substitutes each of the three methods for WPD in the first stage alone, keeping the node-wise (or mode-wise) Wasserstein distance metric, Otsu thresholding and second-stage soft-mask refinement exactly as in the main text, and reports all results with the indices already established there (number of common/differential nodes, relative reconstruction error, CC distance reduction, and DiC distance retention).

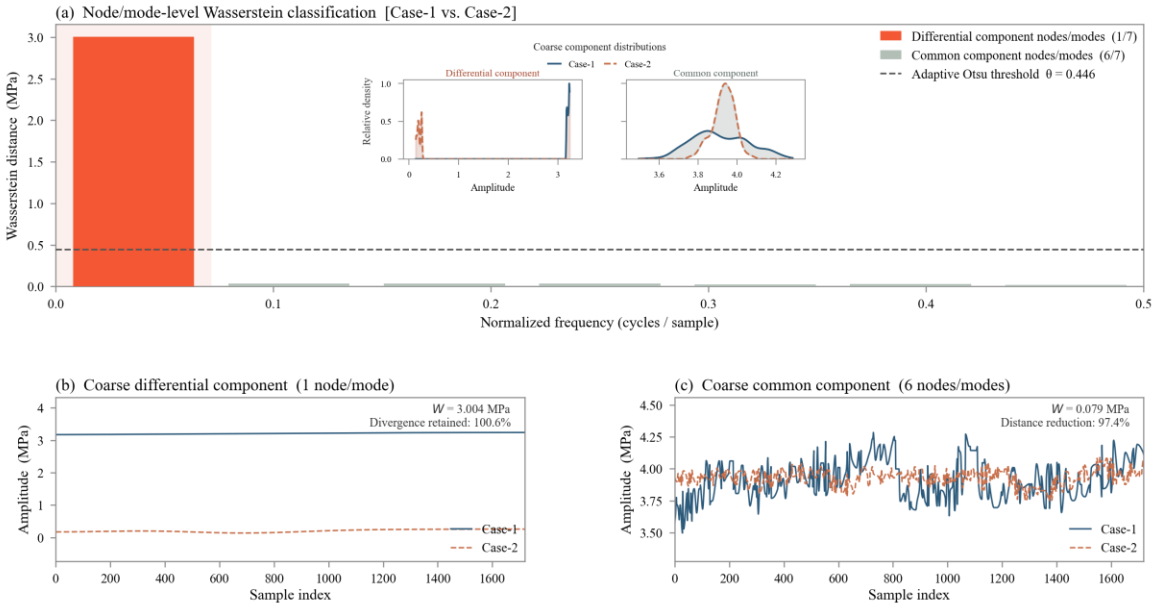
The substitution is admissible because the second-stage CWT–FQI projection and Sigmoid soft-mask refinement act directly on the time-domain common and differential signals output by the first stage and do not depend on the specific sub-band structure of the WPD nodes; they can therefore be applied unchanged to the common and differential signals produced by an EMD/EEMD/VMD first stage. The three baselines are configured as follows. EMD (empirical mode decomposition, Huang et al., 1998) adaptively decomposes a signal into intrinsic mode functions (IMFs) by iterative sifting; it requires no preset parameters but suffers from mode mixing. EEMD (ensemble empirical mode decomposition, Wu and Huang, 2009) takes the ensemble average of EMD results over repeated noise-assisted trials; 30 trials and a noise amplitude of 0.05 times the signal standard deviation are used here. VMD (variational mode decomposition, Dragomiretskiy and Zosso, 2014) casts the decomposition as a constrained variational optimization problem and requires the number of modes to be specified in advance; $K = 5$ and $\alpha = 2000$ are used here. The mode counts of EMD and EEMD are determined by the signal itself, so the two boreholes generally yield different native counts when decomposed independently; this appendix aligns them by index truncation (taking the smaller count and merging the surplus modes into the last one so that the total is preserved). The VMD mode count is fixed, so the two boreholes correspond directly by index.

1.2 Comparability prerequisites of the first-stage coarse decomposition and a first comparison of separation performance

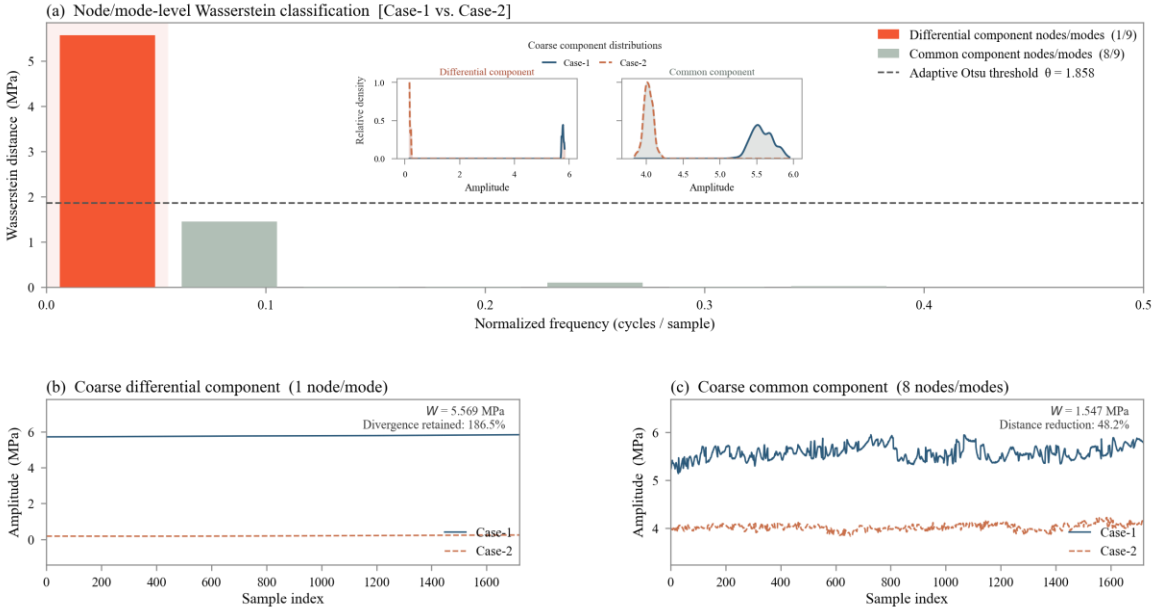
This section compares the first-stage coarse decomposition itself for the four methods. The case is the single-factor rotational-speed pair already used in Sections 2.2.1 and 2.2.2 of the main text (Case-1 vs Case-2, pressure signal); results are given in Fig. 1 and Table 1.


(a) Node/mode-level Wasserstein classification [Case-1 vs. Case-2]
Wasserstein distance (MPa)
Normalized frequency (cycles / sample)
Differential component nodes/modes (1/32)
Common component nodes/modes (31/32)
Adaptive Otsu threshold θ = 0.094
Coarse component distributions
Case-1
Case-2
Differential component
Common component
Relative density
Amplitude
(b) Coarse differential component (1 node/mode)
W = 2.986 MPa
Divergence retained: 100.0%
Amplitude (MPa)
Sample index
(c) Coarse common component (31 nodes/modes)
W = 0.035 MPa
Distance reduction: 98.8%

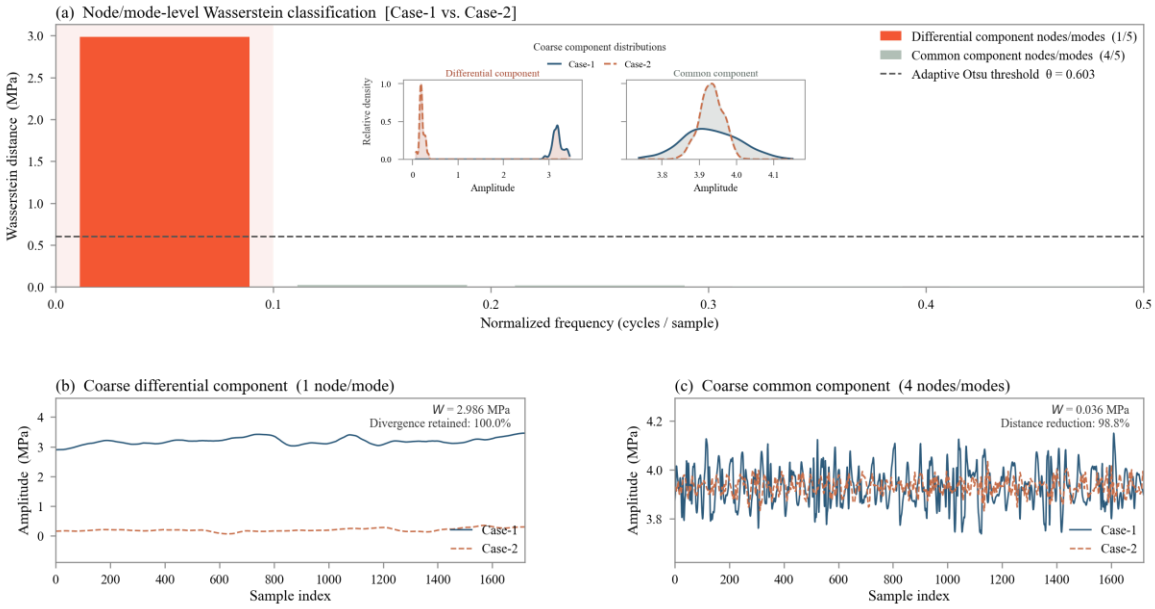
(a) WPD


(a) Node/mode-level Wasserstein classification [Case-1 vs. Case-2]
Wasserstein distance (MPa)
Normalized frequency (cycles / sample)
Differential component nodes/modes (1/7)
Common component nodes/modes (6/7)
Adaptive Otsu threshold θ = 0.446
Coarse component distributions
Case-1
Case-2
Differential component
Common component
Relative density
Amplitude
(b) Coarse differential component (1 node/mode)
W = 3.004 MPa
Divergence retained: 100.6%
Amplitude (MPa)
Sample index
(c) Coarse common component (6 nodes/modes)
W = 0.079 MPa
Distance reduction: 97.4%

(b) EMD



(c) EEMD



(d) VMD (K=5)

Fig. 1. Node/mode classification by Wasserstein distance and the resulting common- and differential-component waveforms for WPD and EMD/EEMD/VMD (Case-1 vs Case-2, pressure signal).

Table 1. Key indices of the four decomposition methods (Case-1 vs Case-2, pressure signal).

Method	Nodes / modes	Common / differential	Otsu threshold	Relative reconstruction error (A / B)	Coarse CC distance reduction	Coarse DiC distance retention
WPD-db4 (level=5)	32	31/1	0.0937	4.50e-16 / 4.70e-16	98.8%	100.0%
EMD (native modes: A =	7	6/1	0.4455	0.00e+00 / 1.11e-17	97.4%	100.6%

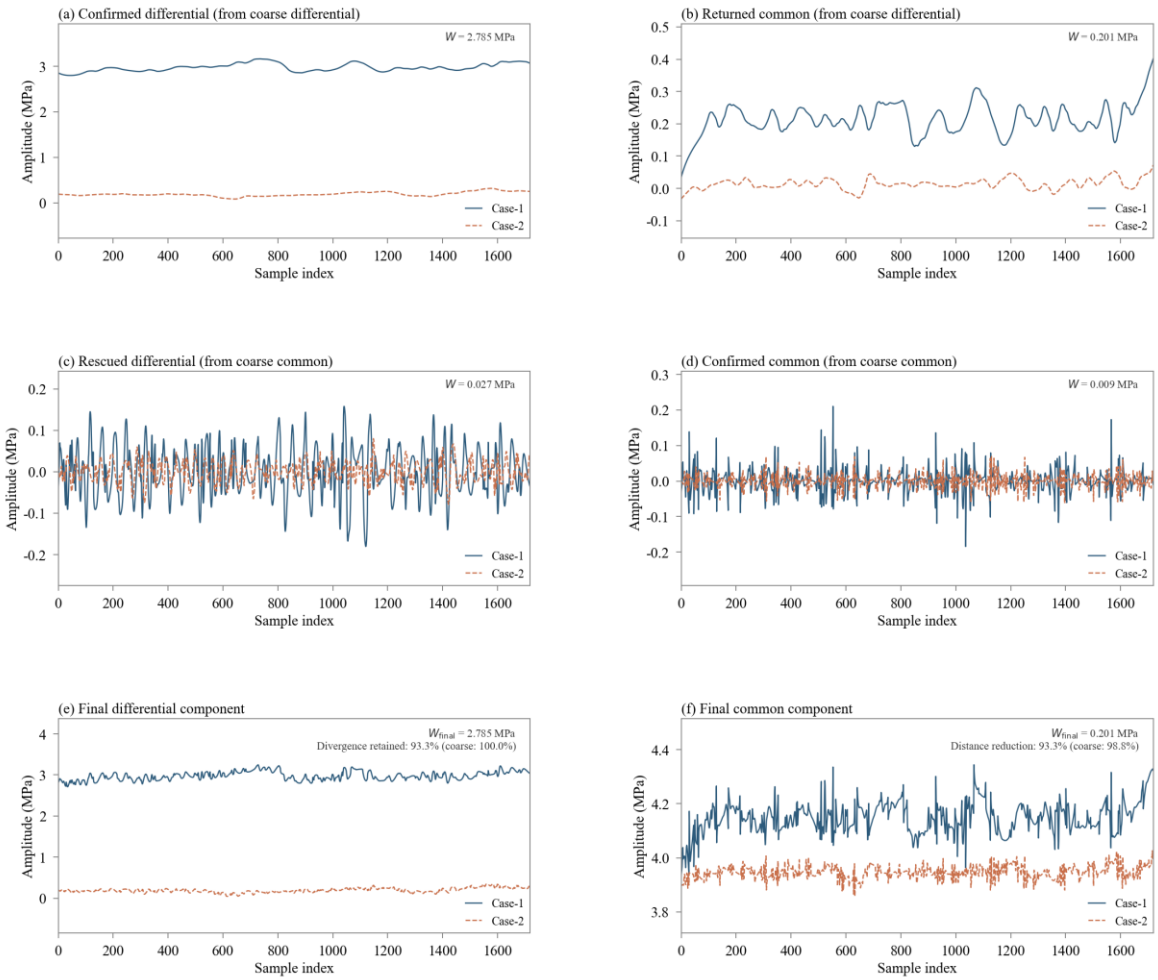
7, B = 8; 7 after alignment)						
EEMD (native modes: A = 10, B = 9; 9 after alignment)	9	8/1	1.8584	1.32e+00 / 3.94e-01	48.2%	186.5%
VMD (K = 5, must be set manually)	5	4/1	0.6033	6.79e-03 / 6.44e-02	98.8%	100.0%

Consider first whether the four methods meet the comparability prerequisites of the framework. The relative reconstruction errors of WPD, EMD and VMD are all at most 6.4×10^{-2} (WPD and EMD reach the machine-precision order of 10^{-16}), so exact reconstruction holds; EEMD, by contrast, reaches 0.39–1.32 (that is, 39%–132%), its component sum departing markedly from the original signal, which cannot support either the four-component merging or the step of adding the baseline back to the common component only. Second, the node/mode-count column of Table 1 reveals a structural difference in cross-borehole comparability: at a fixed number of levels WPD yields 32 dyadic sub-band leaf nodes for any signal, so the node correspondence between the two boreholes is inherently consistent; the mode counts of EMD and EEMD are set by the signal itself (7 and 8, and 10 and 9, for the two boreholes respectively), and a correspondence can be established only after truncation alignment; VMD is fixed at 5, but that value must be specified manually. Because the discrimination mechanism of this paper rests on computing the Wasserstein distance between the two boreholes within the same sub-band node by node, a consistent node correspondence is precisely the condition under which that mechanism can be defined at all.

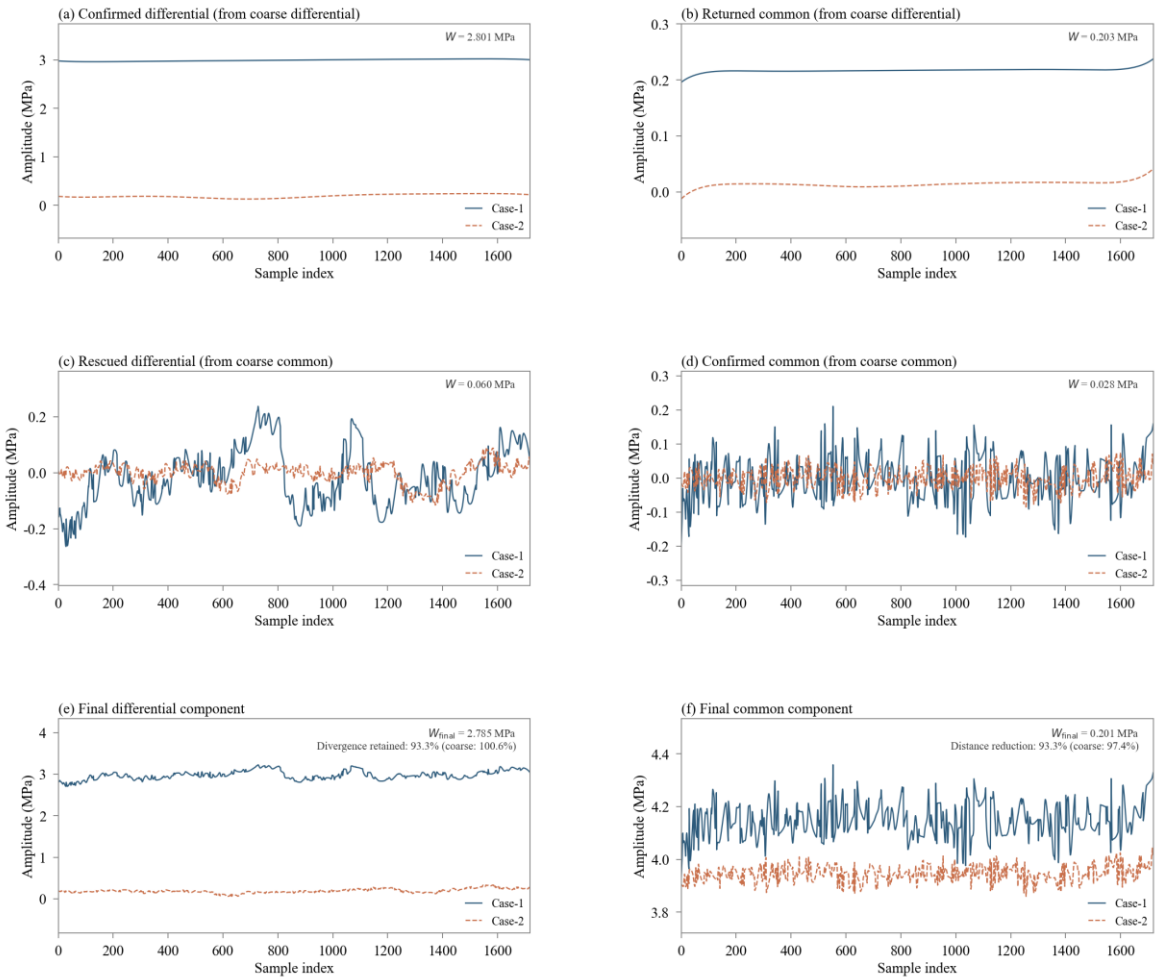
Turning to the first-stage separation itself, the coarse CC distance reductions of WPD, EMD and VMD are 98.8%, 97.4% and 98.8% respectively, whereas EEMD attains only 48.2%. The coarse DiC distance retention of EEMD reaches 186.5% (above 100%, meaning that the inter-borehole distance of the coarse differential component already exceeds that of the original signal), consistent with its failure to meet the exact-reconstruction prerequisite and indicating that its common/differential separation has lost physical meaning.

1.3 Final separation of the complete two-stage workflow (single case)

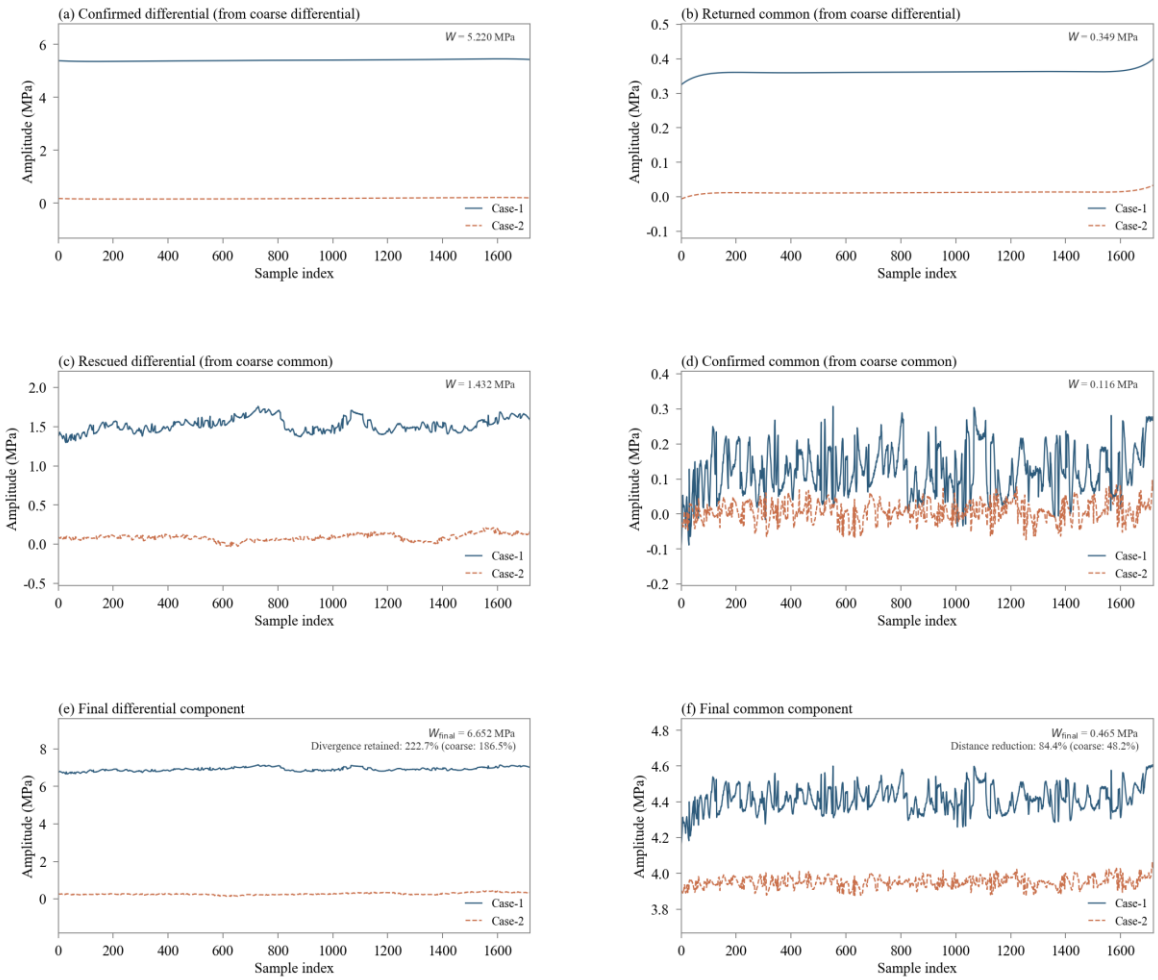
This section applies the complete two-stage workflow to all four methods, so as to examine whether the second-stage refinement can absorb the differences among the first-stage methods. The four sub-components (confirmed differential, returned common, rescued differential and confirmed common) and the final results are shown in Fig. 2 and Table 2.


(a) Confirmed differential (from coarse differential)
W = 2.785 MPa
(b) Returned common (from coarse differential)
W = 0.201 MPa
(c) Rescued differential (from coarse common)
W = 0.027 MPa
(d) Confirmed common (from coarse common)
W = 0.009 MPa
(e) Final differential component
W_final = 2.785 MPa
Divergence retained: 93.3% (coarse: 100.0%)
(f) Final common component
W_final = 0.201 MPa
Distance reduction: 93.3% (coarse: 98.8%)
Amplitude (MPa)
Sample index
Case-1
Case-2

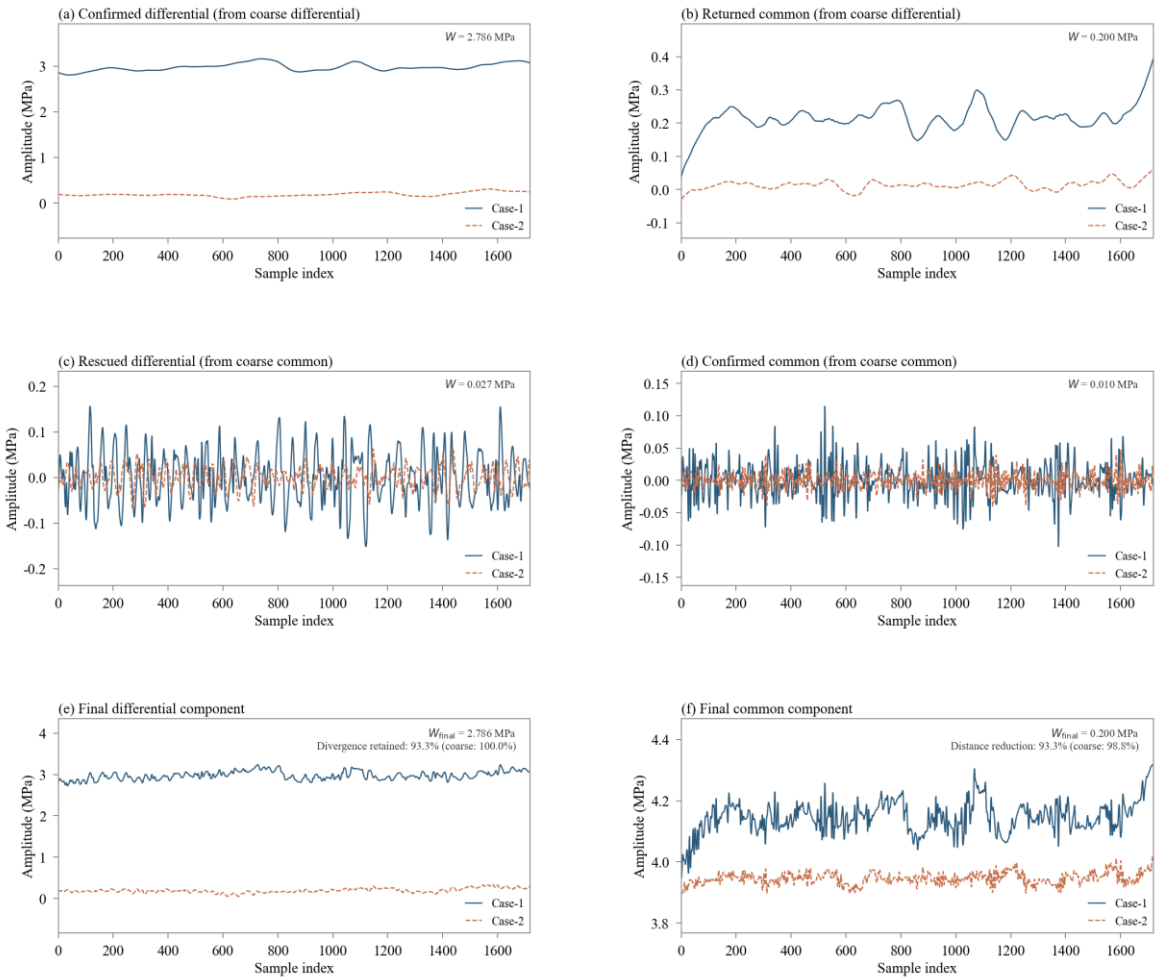
(a) WPD


(a) Confirmed differential (from coarse differential)
W = 2.801 MPa
(b) Returned common (from coarse differential)
W = 0.203 MPa
(c) Rescued differential (from coarse common)
W = 0.060 MPa
(d) Confirmed common (from coarse common)
W = 0.028 MPa
(e) Final differential component
W_final = 2.785 MPa
Divergence retained: 93.3% (coarse: 100.6%)
(f) Final common component
W_final = 0.201 MPa
Distance reduction: 93.3% (coarse: 97.4%)
Amplitude (MPa)
Sample index
Case-1
Case-2

(b) EMD


(a) Confirmed differential (from coarse differential)
W = 5.220 MPa
(b) Returned common (from coarse differential)
W = 0.349 MPa
(c) Rescued differential (from coarse common)
W = 1.432 MPa
(d) Confirmed common (from coarse common)
W = 0.116 MPa
(e) Final differential component
W_final = 6.652 MPa
Divergence retained: 222.7% (coarse: 186.5%)
(f) Final common component
W_final = 0.465 MPa
Distance reduction: 84.4% (coarse: 48.2%)
Amplitude (MPa)
Sample index
Case-1
Case-2

(c) EEMD



(d) VMD (K=5)

Fig. 2. Four sub-components of the two-stage workflow for WPD and EMD/EEMD/VMD (Case-1 vs Case-2, pressure signal).

Table 2. Final DiC retention and CC reduction of the four-sub-component workflow (Case-1 vs Case-2, pressure signal).

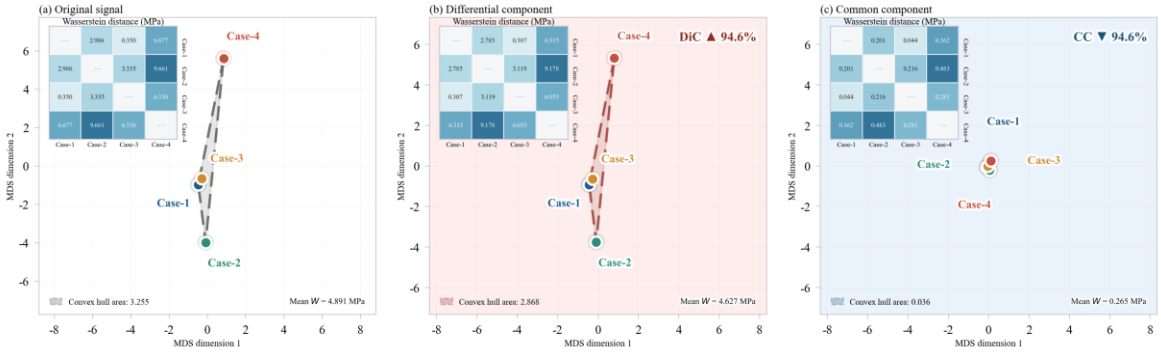
Method	Final DiC retention	Final CC reduction	(reference: first-stage coarse components)
WPD	93.3%	93.3%	Coarse DiC 100.0% / coarse CC 98.8%
EMD	93.3%	93.3%	Coarse DiC 100.6% / coarse CC 97.4%
EEMD	222.7%	84.4%	Coarse DiC 186.5% / coarse CC 48.2%
VMD (K=5)	93.3%	93.3%	Coarse DiC 100.0% / coarse CC 98.8%

After the complete two-stage workflow, the final DiC retention and CC reduction of WPD, EMD and VMD are closely comparable (93.3%, 93.3% and 93.3%), showing that the second-stage soft-mask refinement can correct, to a degree, the bias introduced by differences among the first-stage decomposition methods. The final DiC retention of EEMD, however, rises to 222.7% (far above 100%, that is, after the two-stage workflow the inter-borehole difference of the differential

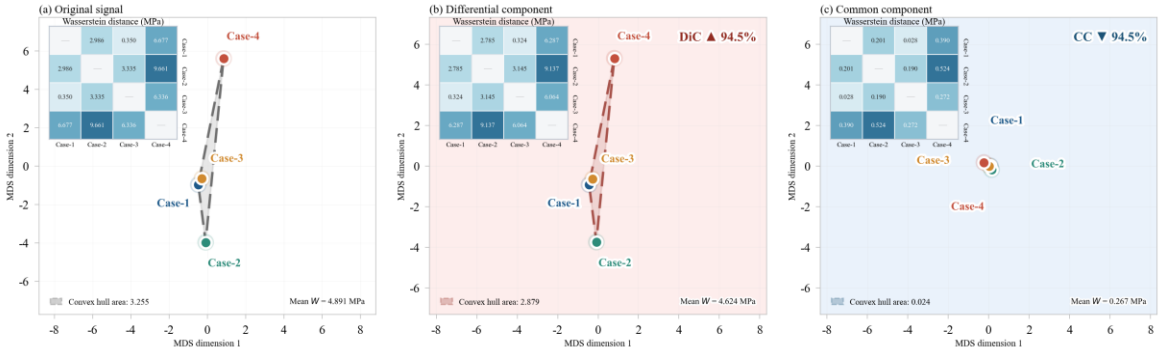
component has been amplified beyond that of the original signal), showing that when the first-stage decomposition itself does not conserve energy, as with EEMD, the second-stage refinement cannot compensate for that defect and may instead amplify the error.

1.4 Generality check over all six borehole pairs

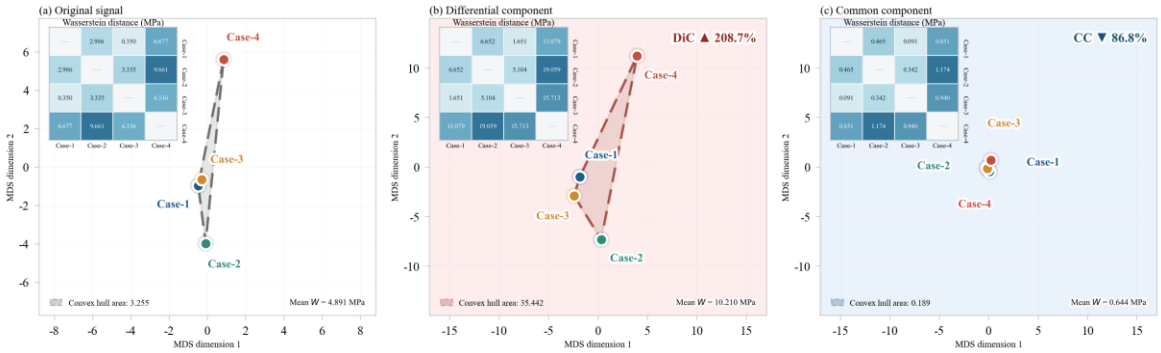
This section applies each of the four methods to all four boreholes of the main dataset of the main text (Case-1 to Case-4, six pairs in total). The original signal, differential component and common component are projected onto a plane by MDS and their convex hulls drawn, so that convergence of the common component and continued divergence of the differential component can be read directly; see Fig. 3 and Table 3.



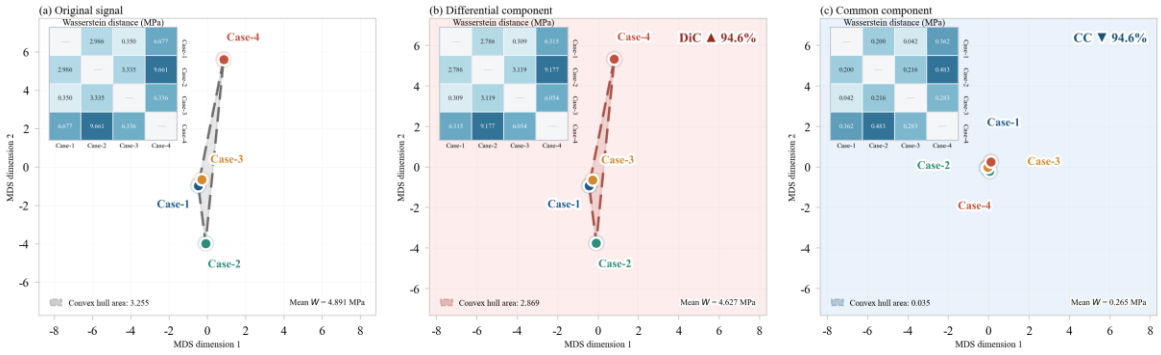
(a) WPD



(b) EMD



(c) EEMD



(d) VMD (K=5)

Fig. 3. Wasserstein spatial structure over four boreholes for WPD and EMD/EEMD/VMD (Case-1 to Case-4, pressure signal, MDS projection).

Table 3. Key indices of the Wasserstein spatial structure over four boreholes (six pairs) (means, pressure signal)

Method	CC distance reduction	DiC distance retention	CC convex hull area
WPD	94.6%	94.6%	0.036
EMD	94.5%	94.5%	0.024
EEMD	86.8%	208.7%	0.189
VMD (K=5)	94.6%	94.6%	0.035

The six-pair means agree with the single case. The CC distance reduction and DiC distance retention of WPD, EMD and VMD all lie close to 94.5%–94.6% (94.6%, 94.5% and 94.6% respectively), and the convex-hull structures of their original signals, differential components and common components (groups (a), (b) and (d) of Fig. 3) are visually almost identical. For EEMD the CC distance reduction falls to 86.8% and the DiC distance retention rises to 208.7%, its differential-component convex hull area (35.442) even exceeding that of the original signal (3.255), so EEMD again fails to separate the common and differential components effectively under the complete two-stage workflow.

1.5 Conclusions

Under the complete two-stage workflow the final common/differential separation of WPD, EMD and VMD is closely comparable (93.3% DiC retention and CC reduction for the single case; 94.5%–94.6% for the six-pair means). This is itself positive evidence for the two-stage funnel-shaped framework: differences among first-stage decomposition methods are absorbed by the second-stage soft-mask refinement, so the separation performance of the framework does not depend on any particular first-stage decomposer and is structurally robust.

On that basis, WPD is chosen for the first stage in the main text because of three structural advantages in the general setting of multiple lithologies, devices and control parameters. (1) Orthogonal wavelet packet decomposition guarantees that the common and differential components sum exactly to the original signal, which is the structural prerequisite for the four-component merging and for adding the baseline back to the common component only; the 39%–132% relative reconstruction error of EEMD makes both steps structurally unrealizable, and its final DiC retention of 222.7% (208.7% for the six-pair mean) reflects not poorer separation quality but a failure to meet that prerequisite. (2) The decomposition depends on no hyperparameter specified in advance, requiring neither a preset mode count K as in VMD nor the mode truncation alignment needed for EMD/EEMD. (3) At a given decomposition level the node structure is fixed, so different boreholes

and different signals share a consistent node correspondence by construction, which is precisely the condition under which the node-wise Wasserstein discrimination mechanism can be defined.